

Yifei Zhang

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Research Interests: Autonomous LLM Agents | Multi-Agent Systems | LLM Applications in Finance

Education

B.Sc. in Computer Science and Technology (FinTech) <i>Department of Computer Science and Technology, Top Talent Class</i>	Nanjing University <i>Sep 2020 – Jun 2024</i>
M.Sc. in Management Science and Engineering <i>School of Management & Engineering, supervised by Prof. Honghai Yu</i>	Nanjing University <i>Sep 2024 – Jun 2027</i>

Intern Experience

ML Group, Microsoft Research Asia (MSRA) <i>Research intern advised by Weiqing Liu and Xiao Yang</i>	<i>May 2025 – Jan 2026</i>
School of Data Science, CUHK-Shenzhen <i>Research intern advised by Prof. Benyou Wang</i>	<i>Jun 2024 – Mar 2025</i>

Research & Projects

FT-Dojo (RD-Agent 12,000+ Stars) <i>Advised by Weiqing Liu and Xiao Yang</i>	<i>Sep 2025 – Jan 2026</i> <i>First Author</i>
* Built the first interactive benchmark for autonomous LLM fine-tuning, covering 13 tasks across 5 domains (math, chemistry, finance, patents, table QA), where agents must complete a full loop of data curation → training → evaluation → iterative diagnosis. * Systematically characterized three core challenges: context explosion, inefficient exploration from lack of fail-fast capability, and limited comprehension of multi-level evaluation feedback. The proposed FT-Agent achieves SOTA on 10 of 13 tasks, outperforming Claude Code, Codex, and other strong baselines. (<i>Under review at ICML 2026; received positive reviews.</i>)	
Gome (RD-Agent 12,000+ Stars) <i>Advised by Weiqing Liu and Xiao Yang</i>	<i>May 2025 – Sep 2025</i> <i>First Author</i>
* Observed that existing MLE agents, despite varied search topologies, uniformly compress rich execution signals (error traces, training logs, code diffs) into scalar scores for candidate <i>ranking</i>, discarding directional update information. Inspired by gradient-based optimization, Gome repurposes feedback from <i>ranking</i> to <i>updating</i>: the LLM performs structured reasoning over execution signals to guide code revision direction (analogous to gradient), with experience memory for faster convergence (momentum) and multi-track parallelism for broader search coverage (distributed SGD). * Achieved 35.1% any-medal rate (open-source SOTA) on MLE-Bench closed-world protocol. Across 10 models (GPT-4o-mini → GPT-5), a crossover phenomenon is observed: tree search dominates on weaker models while Gome significantly outperforms on stronger models (+7.1% on GPT-5), revealing distinct scaling laws for the two paradigms. (<i>ACL Findings 2026</i>)	
TwinMarket <i>Advised by Prof. Benyou Wang, Prof. Honghai Yu</i>	<i>Oct 2024 – Mar 2025</i> <i>First Author</i>
* Led development of TwinMarket, a large-scale LLM-driven multi-agent simulation platform for financial markets. Each agent implements a BDI cognitive architecture with perception-reasoning-action loops, supporting scalable concurrent simulation of 1,000+ agents. * Designed inter-agent social network communication; the system spontaneously exhibits emergent collective intelligence including information cascades and herding effects, replicating statistical stylized facts (fat-tailed returns, volatility clustering). Invited oral presentation at ICLR 2025 Financial AI Workshop; awarded Best Paper (1/53). (<i>NeurIPS 2025</i>)	
UCFE Benchmark <i>Advised by Prof. Benyou Wang, Prof. Honghai Yu, and Jimin Huang</i>	<i>Aug 2024 – Jan 2025</i> <i>Project Leader</i>
* Designed UCFE Benchmark, a user-centric financial expertise evaluation framework. Conducted a survey of 804 participants and constructed a dataset with 330 entries across 17 task types.	

- * Achieved strong human alignment (Pearson **0.78**) using a **GPT-4o** simulator and **LLM-as-Judge**, providing a reproducible automated evaluation pipeline for financial LLM assessment. (*NAACL Findings 2025*)

AI-Trader (12,000+ GitHub Stars)

Nov 2025

Collaboration with *HKUDS Lab, HKU*

A-Share Module Lead

- * The first fully automated, real-time, contamination-free benchmark for LLM-based trading agents. Led construction of the **A-share market module**, implementing T+1 settlement, $\pm 10\%$ price limits, and other China-specific market rules, covering SSE 50 constituents.

Publications (* denotes equal contribution)

UCFE: A User-Centric Financial Expertise Benchmark for Large Language Models

Yuzhe Yang*, **Yifei Zhang***, Yan Hu*, Yilin Guo, Ruoli Gan, Yueru He, Mingcong Lei, Xiao Zhang, Haining Wang, Qianqian Xie, Jimin Huang, Honghai Yu, Benyou Wang.

NAACL Findings 2025

TwinMarket: A Scalable Behavioral and Social Simulation for Financial Markets

Yuzhe Yang*, **Yifei Zhang***, Minghao Wu*, Kaidi Zhang, Yunmiao Zhang, Honghai Yu, Yan Hu, Benyou Wang.

NeurIPS 2025 & **Best Paper Award** at Financial AI @ ICLR 2025 (1/53)

Reasoning as Gradient: Scaling MLE Agents Beyond Tree Search

Yifei Zhang*, Xu Yang*, Xiao Yang*, Bowen Xian, Qizheng Li, Shikai Fang, Jingyuan Li, Jian Wang, Mingrui Xu, Weiqing Liu, Jiang Bian.

ACL Findings 2026

FT-Dojo: Towards Autonomous LLM Fine-Tuning with Language Agents

Qizheng Li*, **Yifei Zhang***, Xiao Yang*, Xu Yang*, Zhuo Wang, Weiqing Liu, Jiang Bian.

Under review at ICML 2026. Received positive reviews.

Honors & Awards

China National Scholarship for Master's Students, Nanjing University (**top 0.2%**)

Oct 2025

National Gold Award, China Int'l College Students Innovation Competition (Youth Red Track)

Oct 2025

Outstanding Graduate Student Model, Nanjing University

Nov 2025